

DEEP LEARNING-BASED DEEPPFAKE DETECTION FOR SOCIAL MEDIA CONTENT

Expanded Technical Assignment Report

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1. Executive Summary

Deepfake media has become an important artificial-intelligence application problem because modern generative methods can create highly realistic facial manipulations. Social-media platforms accelerate the creation, compression, re-encoding and redistribution of such content. The assignment therefore frames deepfake detection as a practical screening problem rather than as a claim of absolute forensic certainty.

The proposed system accepts an image or short video, validates the input, samples video frames when necessary, detects visible faces, crops and normalizes the facial regions, and applies a transfer-learning-based convolutional neural network for binary classification. For video, frame-level scores are aggregated to obtain a video-level score and final decision.

The source assignment specifies representative prototype results of 94.20% accuracy, 93.80% precision, 95.10% recall, 94.44% F1-score and 0.97 ROC-AUC, with approximately 0.21 seconds average image inference time and 92.50% video-level accuracy. These figures are treated here as the representative results supplied in the assignment, not as universal forensic performance.

The engineering rationale is centered on practicality. Transfer learning reduces training requirements compared with building a large CNN from scratch. Face-focused preprocessing reduces irrelevant background information, while configurable frame sampling controls the computational cost of video analysis. Validation-based threshold selection is preferred to blindly assuming a 0.5 threshold.

2. Introduction and Background

Artificial intelligence has transformed image and video generation. The same progress that enables useful creative applications also enables manipulated facial media, including face swaps, facial reenactments and synthetic identities. When such content is distributed through social platforms, viewers may encounter it without knowing whether the underlying media is authentic.

Deepfake detection is therefore an applied computer-vision task. The system must learn visual patterns associated with genuine and manipulated facial content while coping with variations introduced by cameras, lighting, pose, resolution, compression, resizing, cropping and re-encoding.

The assignment positions the detector as a preliminary screening layer. Its output is a probability or confidence-like score that supports review. This distinction is important because a machine-learning score is not equivalent to absolute truth. A robust engineering report must therefore discuss uncertainty, false positives, false negatives and generalization.

The proposed workflow connects several Artificial Intelligence course concepts: machine-learning pipelines, computer vision, convolutional neural networks, transfer learning, data augmentation, model evaluation and error analysis. It also demonstrates how an academic model can be converted into a practical application handling both valid and invalid media inputs.

3. Problem Understanding

The central problem is to distinguish REAL facial media from DEEPFAKE facial media. For images, the task is a binary classification problem over an input facial region. For videos, the problem is extended by sampling multiple frames and combining the resulting predictions.

Input media may differ substantially in visual quality. A high-quality original photograph and a compressed social-media image can contain the same identity but produce different pixel distributions. Similarly, a manipulated video may contain only some frames with strong detectable artifacts. This motivates frame sampling and aggregation rather than relying on one arbitrary video frame.

The expected output contains three practical components: a predicted class, a score or confidence value, and—when the input is a video—an aggregated video-level result. Optional frame evidence can help a reviewer understand why the system reached its decision.

The system assumptions include reliable labels, detectable faces for most samples, and strict separation between training and testing data. The assignment explicitly warns that identities or source videos must not leak across partitions, because leakage could make the evaluation appear stronger than genuine generalization.

4. Problem Statement and Objectives

The design challenge is to implement a deep learning-based deepfake detection system for social-media-style image and video content. The system should validate input media, extract or sample frames, identify relevant facial regions, classify the content, aggregate video predictions, report evaluation metrics and handle failures safely.

The first objective is reliable media handling. Supported JPEG/PNG-style images and MP4/MOV-style videos should be processed, while unsupported, corrupted or unreadable inputs should generate clear errors instead of misleading predictions.

The second objective is efficient visual analysis. The system should focus on visible faces, resize crops to the model's required input size, normalize pixels consistently, and use training-only augmentation to improve robustness to common variations.

The third objective is measurable performance. Accuracy, precision, recall, F1-score, ROC-AUC, inference time and video-level accuracy should be considered. The system should also test transformations such as compression, resizing and blur where applicable.

The final objective is responsible deployment. The detector must be described as a screening aid, with explicit attention to privacy, fairness, reliability, security and the consequences of incorrect decisions.

5. Deepfake Types and Threat Landscape

The source assignment identifies several forms of manipulated media: face swaps, facial reenactments, synthetic identities and other deepfake content. These categories differ in how the face is generated or altered, but they share the practical challenge of producing media that may look authentic to human viewers.

Face swapping replaces or synthesizes a subject's facial appearance while retaining much of the original scene. Facial reenactment attempts to transfer expressions or movements. Synthetic identities can represent faces that do not correspond to a real person. Each manipulation type may leave different visual traces, and a detector trained on one family may not automatically generalize to another.

Social-media processing adds another layer of difficulty. Compression can remove high-frequency details, resizing can change texture information, cropping can remove contextual evidence, and repeated encoding can introduce artifacts unrelated to the manipulation itself.

The threat is not only technical. Manipulated media can be used for misinformation, impersonation, fraud, harassment or manipulation of public opinion. Therefore, detection systems have a societal role, but their outputs must be handled cautiously because incorrect accusations can also cause harm.

6. Dataset and Input Specification

The assignment specifies images in JPEG/PNG or equivalent supported formats and short videos in formats such as MP4 and MOV. The core labels are binary: REAL and DEEPFAKE. A frame score is represented on a 0.00–1.00 scale, and a video score is obtained by aggregating frame-level scores.

A useful dataset must contain both genuine and manipulated examples. More importantly, partitioning must occur at the identity or original-video level. If frames from one video appear in both training and testing, the model can partially memorize source-specific information rather than learning manipulation-related patterns.

The preprocessing specification includes configurable frame sampling, face detection, fixed-size resizing and model-compatible normalization. Training augmentation may include horizontal flipping, mild blur or compression, brightness variation and controlled cropping.

Dataset documentation should include source, labels, licence or usage conditions, split strategy and any transformations applied. The original assignment requires that actual experimental performance be reported and explicitly prohibits fabricated results.

7. Data Preparation and Leakage Prevention

Data preparation is one of the most important stages because a detector can appear highly accurate under an invalid evaluation design. The safest approach is to group related frames and identities before creating train, validation and test partitions.

For video datasets, all sampled frames originating from the same source video should remain in one partition. Where multiple manipulated versions are derived from the same original identity or source, the grouping policy should also be considered. This reduces the chance that near-duplicate visual information crosses the partition boundary.

The validation set is used for decisions such as threshold selection and model selection. The test set should remain untouched until the final evaluation. This separation provides a cleaner estimate of how the finished system performs on previously unseen samples.

Preprocessing must also be consistent. The same resizing and normalization rules used during validation and testing should match the model's expected input. Augmentation should be applied during training only unless a specific robustness experiment intentionally modifies the evaluation input.

8. System Requirements and Constraints

Functional requirements include image and video input handling, media-format and file-size validation, frame sampling, face detection, cropping, normalization, model inference, binary classification, video aggregation, metric calculation, error reporting and result visualization.

The suggested implementation environment is Python with PyTorch or TensorFlow for deep learning, OpenCV for image and video processing, NumPy/Pandas for data handling, scikit-learn for evaluation and Matplotlib for plots. Git and GitHub are suggested for version control and submission.

Important constraints include documentation of dataset licence conditions, avoidance of identity or video leakage, testing with genuine and manipulated inputs, actual execution evidence, explanation of limitations and a design that can later accept additional manipulation types and datasets.

Compute is also a practical constraint. A student-scale project should avoid unnecessary processing of every frame in long videos. Sampling provides a controllable compromise between temporal coverage and inference cost.

9. Proposed System Architecture

The complete pipeline is: Social Media Image/Video → Media Validation → Frame Sampling → Face Detection & Cropping → Preprocessing → Deep Learning Classifier → Frame-Level Scores → Video Score Aggregation → Decision & Confidence → Detection Result.

Media validation forms the first safety boundary. Invalid extensions, unreadable files and unsupported content should be rejected. For a valid image, the pipeline can proceed directly to face detection. For a valid video, a configurable subset of frames is first selected.

Face detection focuses the classifier on the relevant facial evidence. After cropping, the face is resized and normalized. The deep-learning model then produces a binary prediction or score. For video, multiple frame scores are combined and compared with a validation-selected threshold.

The architecture is modular. Each stage can be replaced independently—for example, a different face detector, a different CNN backbone or a different aggregation method—without redesigning the entire application.

Stage	Main operation	Purpose
1	Media validation	Check extension, readability and supported format
2	Frame sampling	Reduce video computation
3	Face detection	Locate relevant facial regions
4	Face crop	Remove irrelevant scene information
5	Resize & normalize	Create model-compatible input
6	CNN classifier	Predict REAL/DEEPFAKE score
7	Aggregation	Combine frame-level scores
8	Decision	Apply validation-selected threshold
9	Result	Display class, score and optional evidence

Figure/Table: End-to-end processing stages.

10. Image Processing Pipeline

Image inference begins with file validation and decoding. The system should confirm that the file can actually be opened rather than trusting the extension alone. This prevents corrupted inputs from entering the model pipeline.

Once decoded, the image is passed to a face detector. If one or more relevant faces are found, the detector produces facial crops. These crops are resized to the dimensions required by the classifier and normalized using the model-compatible preprocessing procedure.

The classifier produces a score representing the likelihood associated with the DEEPFAKE class. The application then applies the selected threshold to produce the final REAL or DEEPFAKE label. The score should be displayed alongside the label so the user can see that the result is probabilistic.

If no face is detected, the application should report 'No Face Detected' rather than silently returning a classification based on irrelevant background content. Very low-quality images may still be processed with a warning, depending on implementation policy.

11. Video Processing Pipeline

Video processing extends the image pipeline over time. Instead of running the classifier on every frame, the system samples a configurable number of frames. Sampling reduces redundant computation while preserving enough temporal coverage for a preliminary screening result.

For each usable frame, the system detects a face, crops the relevant region, performs the same resize and normalization steps used for images, and obtains a frame-level score. Frames where no face is available can be skipped or reported according to the implementation policy.

The source assignment gives a simple mean aggregation: $\text{Video Score} = (p_1 + p_2 + \dots + p_n) / n$. This makes the video result easy to explain and implement. Alternative methods such as median, trimmed mean and confidence-weighted aggregation can be investigated later.

A video with too few usable frames should not be treated as a normal successful inference. The system should provide an appropriate warning, because a decision based on insufficient visual evidence may be less reliable.

12. Face Detection and Preprocessing

Face detection is used to isolate the region most relevant to the classification task. Background objects, scenery and unrelated image regions can otherwise dominate the learned representation and reduce the model's focus on manipulation cues.

Cropping should preserve enough facial context while avoiding unnecessary background. Resizing converts different source resolutions into a consistent model input. Normalization then maps pixel values into the representation expected by the pretrained backbone.

Training augmentation is designed to expose the model to controlled variations. The assignment proposes horizontal flipping, cropping, mild blur or compression and brightness variation. These transformations should not destroy the manipulation evidence that the detector needs to learn.

Preprocessing is also important for robustness. Social-media content commonly undergoes resizing and compression. By exposing the training process to representative variations and evaluating transformed test samples, the system can reveal how sensitive its learned cues are.

13. Transfer Learning Model

The selected baseline is a transfer-learning-based convolutional neural network. A pretrained visual backbone already contains useful low- and mid-level representations, reducing the amount of training required compared with building a large CNN entirely from scratch.

The classification head is replaced with a binary output. Depending on the implementation, the model may use a two-class output or a single sigmoid output. The exact architecture, input resolution, optimizer, learning rate, batch size, number of epochs and random seed should be recorded in the final implementation.

The assignment identifies Xception, EfficientNet and ResNet as possible alternatives when resources permit. A temporal model is another future option because video contains motion and consistency information that a frame-only classifier does not directly model.

Transfer learning is therefore a practical engineering compromise: it offers a modern deep-learning workflow, manageable compute requirements and a clear experimental baseline.

14. Training Methodology

Training begins with the leakage-free dataset partitions. The model is trained on the training partition while the validation partition is used to monitor performance and guide choices such as the decision threshold.

The optimization configuration must be documented. At minimum, the final report should record the backbone, input dimensions, optimizer, learning rate, batch size, epochs, augmentation policy and random seed. These details make the experiment reproducible and allow meaningful comparisons between runs.

Binary classification performance should be monitored using more than accuracy. Precision, recall and F1-score reveal different failure characteristics, while ROC-AUC measures the model's ranking ability across thresholds.

The training process should be accompanied by actual execution evidence. The assignment specifically recommends screenshots of source code, training or inference execution, successful predictions, error handling, evaluation plots and the final interface.

15. Video Score Aggregation

Video-level classification is obtained by combining frame-level scores. The supplied baseline uses the arithmetic mean. If p_1 through p_n are frame scores, the video score equals their sum divided by the number of usable frames.

Mean aggregation is attractive because it is simple, transparent and computationally inexpensive. However, it assumes that each sampled frame contributes equally. A small number of highly informative frames may be diluted by many neutral frames.

Median aggregation can reduce sensitivity to outlier predictions. A trimmed mean can reduce the influence of extreme values. Confidence-weighted aggregation can give more influence to frames with stronger facial evidence. These alternatives can be compared as future experiments.

The threshold should be selected using validation data rather than assuming a universal 0.5 value. This is particularly important when the costs of false positives and false negatives differ.

16. Evaluation Metrics

Accuracy measures the fraction of all samples classified correctly: $(TP + TN) / (TP + TN + FP + FN)$. It provides a simple overall view but can hide class-specific weaknesses.

Precision is $TP / (TP + FP)$. It describes how many samples predicted as manipulated are actually manipulated and therefore relates to false-alarm control.

Recall is $TP / (TP + FN)$. It measures how effectively the detector finds manipulated samples. A detector with low recall can miss a substantial portion of deepfakes even if its overall accuracy appears reasonable.

F1-score is $2PR / (P + R)$, balancing precision and recall. ROC-AUC represents the area under the receiver operating characteristic curve and measures discrimination across thresholds.

Inference time is also important because a screening system must be computationally practical. For videos, video-level accuracy provides an additional measure beyond individual frame performance.

Metric	Formula / meaning	Purpose
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall correctness
Precision	$TP/(TP+FP)$	False-alarm control
Recall	$TP/(TP+FN)$	Manipulation detection coverage
F1-score	$2PR/(P+R)$	Precision-recall balance
ROC-AUC	Area under ROC	Ranking/discrimination
Inference time	Elapsed time	Computational practicality

Evaluation should be interpreted jointly rather than using accuracy alone.

17. Experimental Test Cases

The assignment provides representative test cases covering normal and failure scenarios. TC1 uses an authentic image and reports REAL with 0.96 confidence. TC2 uses a manipulated image and reports DEEPFAKE with 0.94 confidence.

TC3 uses an authentic video and reports REAL with 0.93 confidence. TC4 uses a manipulated video and reports DEEPFAKE with 0.95 confidence. These cases demonstrate the intended end-to-end operation for both media types.

TC5 covers corrupted or unsupported input and expects an Input Error. TC6 covers a video with no detectable face and reports No Face Detected. These cases demonstrate that safe failure is part of the system design rather than an optional feature.

TC7 evaluates compressed media and reports a DEEPFAKE prediction with 0.89 confidence and 94% robustness in the supplied representative result. TC8 evaluates resized media and reports REAL with 0.91 confidence and 95% robustness.

Test	Input	Expected	Actual / status
TC1	Authentic image	REAL	REAL (0.96), PASS
TC2	Manipulated image	DEEPFAKE	DEEPFAKE (0.94), PASS
TC3	Authentic video	REAL	REAL (0.93), PASS
TC4	Manipulated video	DEEPFAKE	DEEPFAKE (0.95), PASS
TC5	Corrupt/unsupported	Input Error	Rejected, PASS
TC6	No detectable face	Warning / No Face	No Face Detected, PASS
TC7	Compressed media	Prediction + robustness	DEEPFAKE (0.89), 94%, PASS
TC8	Resized media	Prediction + robustness	REAL (0.91), 95%, PASS

Functional and robustness test cases supplied in the source assignment.

18. Results and Validation

The representative performance table in the source assignment reports 94.20% accuracy, 93.80% precision, 95.10% recall, 94.44% F1-score and 0.97 ROC-AUC. Average image inference time is approximately 0.21 seconds per image, while video-level accuracy is 92.50%.

These results indicate a strong binary classification baseline on the evaluated test set. Recall is slightly higher than precision, meaning the supplied run emphasizes detection coverage somewhat more than false-alarm avoidance.

The results must be interpreted within the scope of the evaluated data. The assignment explicitly states that these figures should not be treated as universal forensic accuracy. A detector can perform well on known manipulation methods yet fail on unseen generation techniques.

Validation also includes functional behavior. The supplied test cases show successful predictions for genuine and manipulated images and videos, as well as correct handling of invalid files and no-face inputs.

Metric	Representative value
Accuracy	94.20%
Precision	93.80%
Recall	95.10%
F1-score	94.44%
ROC-AUC	0.97
Average inference time	0.21 seconds/image
Video-level accuracy	92.50%

Representative results as supplied in the original assignment.

19. Robustness to Social-Media Transformations

Social-media transformations are an important source of distribution shift. Compression, resizing and blur can alter the visual signals used by a classifier. A detector that performs well only on clean laboratory-like data may degrade after content passes through a platform's processing pipeline.

The supplied representative results report approximately 94% robustness under the tested compression condition and 95% under resizing. The analysis notes that these results show resilience but also a measurable performance drop.

Robustness testing should compare the original prediction with transformed versions of the same content. This provides a direct view of whether the transformation changes the decision or substantially reduces confidence.

Future experiments could expand the transformation set to include stronger compression, repeated re-encoding, cropping, brightness changes and mild blur. Such tests would help characterize the operating limits of the system.

20. Error Analysis

Error analysis separates false positives and false negatives. A false positive occurs when authentic media is predicted as DEEPFAKE. The assignment suggests reviewing unusual compression, lighting, camera processing and other authentic artifacts that may resemble manipulation cues.

False negatives are especially important because they represent manipulated samples that the detector misses. Possible causes include low-quality manipulation, unseen generation methods, weak facial visibility and transformations that remove detectable cues.

Error analysis should be organized by failure condition rather than by individual examples alone. Useful categories include face size, image quality, lighting, pose, compression level, manipulation type and confidence range.

The purpose of error analysis is not merely to list mistakes. It should guide engineering decisions. If failures cluster around low-resolution faces, preprocessing may need improvement. If failures cluster around unseen manipulation methods, broader training data or stronger generalization methods may be required.

21. Computational Cost and Engineering Trade-offs

A deepfake detector must balance accuracy with computational practicality. Processing every frame of a video can be expensive and often provides redundant information. Frame sampling reduces the number of classifier calls while preserving a representative view of the sequence.

Transfer learning reduces the need to train a large network from scratch. This makes the approach suitable for student-scale compute while still demonstrating modern deep learning. The trade-off is that pretrained features are not specifically designed for forensic manipulation detection.

Temporal deep learning could provide stronger evidence by modeling consistency across frames, but it increases implementation and compute complexity. Multi-modal audio-video analysis can provide additional evidence but requires more data and processing.

The selected design therefore prioritizes a reliable, explainable prototype over maximum possible complexity. This is appropriate for an academic engineering project where the complete pipeline and validation matter as much as model sophistication.

22. Modern Tools and Implementation Structure

The recommended implementation uses Python as the main programming environment. PyTorch or TensorFlow can provide the deep-learning model, OpenCV can handle image and video processing, NumPy and Pandas can support data operations, and scikit-learn can calculate evaluation metrics.

A modular repository makes the system easier to understand and maintain. The assignment suggests files such as `preprocessing.py`, `face_detection.py`, `model.py`, `inference.py`, `video_pipeline.py` and `evaluation.py`, together with an app interface and results directory.

GitHub should maintain source code, configuration, README documentation, evaluation scripts and version history. Sensitive credentials, restricted datasets and passwords should never be committed to the repository.

Execution evidence should correspond to actual runs of the submitted code. This includes screenshots of predictions, evaluation plots, error handling and the final interface. Such evidence strengthens the connection between reported results and the implementation.

23. Security, Privacy, Fairness and Responsible Use

Uploaded media may contain identifiable faces, so privacy must be considered throughout the pipeline. Processing should be controlled and unnecessary retention avoided. Technical logs should not expose sensitive media.

Fairness is another consideration. Dataset composition can affect model behavior, so performance should not automatically be assumed to be uniform across all populations. Evaluation should examine whether certain groups or visual conditions are systematically harder to classify.

Reliability requires clear communication of uncertainty. A confidence score should not be presented as proof of authenticity or manipulation. The detector is best understood as a preliminary screening mechanism that can support human review.

Professional responsibility is especially important because a false deepfake accusation can affect reputation and public trust. The source assignment therefore recommends that the detector not be the sole basis for legal, employment, disciplinary or other high-impact decisions.

Security includes safe file validation and protection of credentials. Since media files can be malformed or unexpectedly large, input validation is both a reliability and security requirement.

24. Limitations and Future Enhancements

The principal limitation is generalization. A detector trained on known manipulation methods may perform poorly on future or unseen generation techniques. High test-set performance therefore cannot be interpreted as a universal guarantee.

The current video approach also uses frame aggregation rather than explicit temporal modeling. This means it does not directly learn motion consistency, identity consistency across frames or temporal artifacts.

Future work can evaluate multiple datasets and unseen manipulation methods. Additional improvements include temporal deep learning, audio-visual analysis, model calibration, stronger robustness testing and a human-review interface.

Deployment optimization is another future direction. A lightweight model could make preliminary screening practical on consumer hardware. Additional manipulation types and datasets should also be supported through the modular pipeline design.

Future evaluation should continue to report actual experimental evidence. New metrics and experiments should be documented clearly so that improvements are distinguishable from changes in dataset composition or evaluation methodology.

25. Conclusion and Student Reflection

The proposed system provides a structured engineering approach for preliminary screening of manipulated images and videos. It combines media validation, frame sampling, face detection, preprocessing, transfer-learning-based classification, video score aggregation and quantitative evaluation.

The principal engineering decision is transfer learning because it balances model capability, training cost and implementation complexity. Face-focused preprocessing reduces irrelevant scene information, while frame aggregation extends image classification to video without requiring a complex temporal architecture.

The representative prototype results supplied in the assignment are 94.20% accuracy, 93.80% precision, 95.10% recall, 94.44% F1-score and ROC-AUC of 0.97, with approximately 0.21 seconds per image and 92.50% video-level accuracy.

The assignment reflection emphasizes learning how a real-world problem becomes an end-to-end machine-learning system. Key lessons include data preparation, leakage-free splitting, preprocessing, transfer learning, evaluation metrics and error analysis. With additional resources, the detector could be evaluated across multiple datasets, unseen methods and richer temporal or multi-modal models.

26. References and Assessment Mapping

The source assignment references FaceForensics++, Celeb-DF, the DeepFake Detection Challenge dataset, ResNet, EfficientNet, and documentation for PyTorch/TensorFlow, OpenCV and scikit-learn. It also states that any additional datasets, software, external sources or AI-assisted tools actually used should be acknowledged according to the institution's required citation format.

The assessment rubric allocates 10 marks to problem understanding and formulation, 20 to application of course/domain knowledge, 20 to solution methodology/design/implementation, 10 to modern tool usage, 15 to results/testing/validation, 15 to analysis/trade-offs/justification, 5 to broader considerations and 5 to technical documentation/reflection.

The CO–PO mapping connects problem formulation with CO1/PO1, implementation with CO2/PO3/PO5, validation with CO2/PO4, analysis with CO2/PO2 and PO4, broader considerations with CO2/PO6 and PO7, and documentation/reflection with CO1/CO2 and PO10.

This expanded report preserves the assignment's central terminology, methodology, representative results and assessment structure while providing additional explanatory depth for a longer technical submission.

Appendix A – Functional Requirements Checklist

The following checklist expands the implementation and documentation requirements already defined in the assignment.

- Image upload and validation
- Video upload and validation
- Configurable frame sampling
- Face detection and crop generation
- Model-compatible resize and normalization
- Deep-learning inference
- Binary REAL/DEEPFAKE prediction
- Video score aggregation
- Threshold selection using validation data
- Metric calculation
- Error handling for corrupted media
- No-face warning
- Robustness testing
- Result visualization
- Execution evidence

Appendix B – Suggested Implementation Configuration Record

The following checklist expands the implementation and documentation requirements already defined in the assignment.

- Dataset name and version
- Dataset source and licence
- Train/validation/test split strategy
- Identity/video leakage prevention method
- Face detector used
- CNN backbone used
- Input resolution
- Optimizer
- Learning rate
- Batch size
- Number of epochs
- Augmentation policy
- Random seed
- Video frame sampling count
- Aggregation method
- Decision threshold
- Hardware used
- Average inference time
- Evaluation date

Appendix C – Edge-Case Handling Matrix

The following checklist expands the implementation and documentation requirements already defined in the assignment.

- Unsupported file extension → Input Error
- Corrupted image → Media Read Error
- Corrupted video → Media Read Error
- No detectable face → No Face Detected
- Too few usable video frames → Warning
- Very low-resolution face → Warning / prediction according to policy
- Compressed readable media → Prediction + robustness measurement
- Resized readable media → Prediction + robustness measurement
- Empty input → Input Required
- Oversized input → Reject or controlled handling

Appendix D – Viva / Presentation Discussion Points

The following checklist expands the implementation and documentation requirements already defined in the assignment.

- Why is deepfake detection a binary classification problem?
- Why is face cropping useful?
- Why is transfer learning suitable for student-scale compute?
- Why must identities and source videos be separated across splits?
- Why is accuracy alone insufficient?
- What is the difference between precision and recall?
- Why is frame sampling used for videos?
- Why should the threshold be selected using validation data?
- What are false positives and false negatives?
- Why is the detector not conclusive forensic evidence?
- What happens when no face is detected?
- How does compression affect the model?
- What would temporal deep learning add?
- How can privacy be protected?
- Why must reported performance come from actual experiments?

Appendix E – Final Submission Checklist

The following checklist expands the implementation and documentation requirements already defined in the assignment.

- Assignment problem statement included
- Objectives explained
- Dataset and licence documented
- Leakage prevention documented
- Methodology and architecture included
- Implementation tools listed
- Training configuration recorded
- Actual execution evidence attached
- Valid inputs tested
- Invalid and edge inputs tested
- Accuracy/precision/recall/F1/ROC-AUC reported
- Inference time reported
- Video-level performance reported
- Robustness testing included
- Error analysis included
- Limitations explained
- Privacy/fairness/security considerations included
- Future work included
- Student reflection included
- References acknowledged